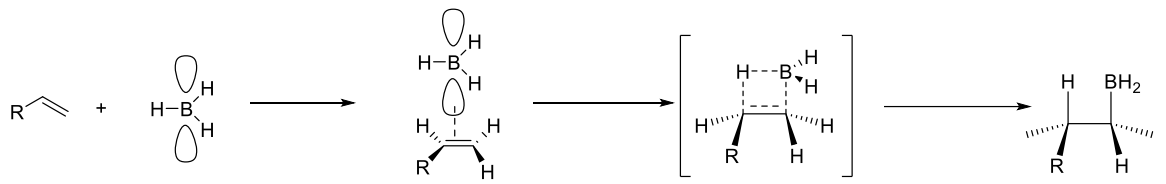
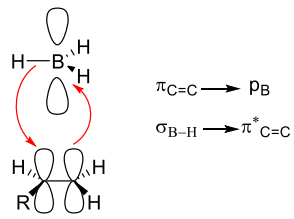
Hydroboration - oxidation reaction :



π – complex formation Concerted T.S Syn - addition

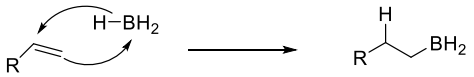
In MO terms, the addition can be viewed as taking place by interaction of the alkene orbital with vacant p - orbital on boron, accompanied by concerted C-H bond formation by interaction with empty π^* orbital of alkene.



Is it pericyclic?

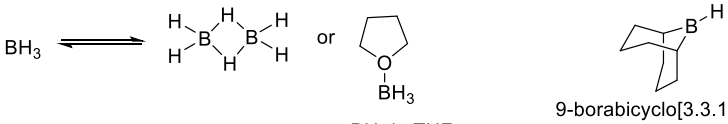
No. The empty p - orbital on boron is the electrophilic site and the σ_{B-H} orbital is the nucleophilic site. These orbitals are orthogonal. The addition is not pericyclic.

Arrow pushing mechanism :



H at the most substituted position.
(Anti - Markonikov)

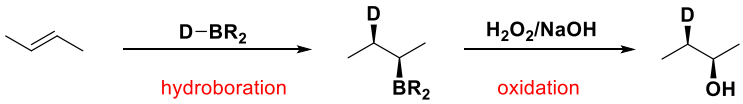
Regioselectivity (Electronics + Sterics): Bulky borane favors anti-Markovnikov



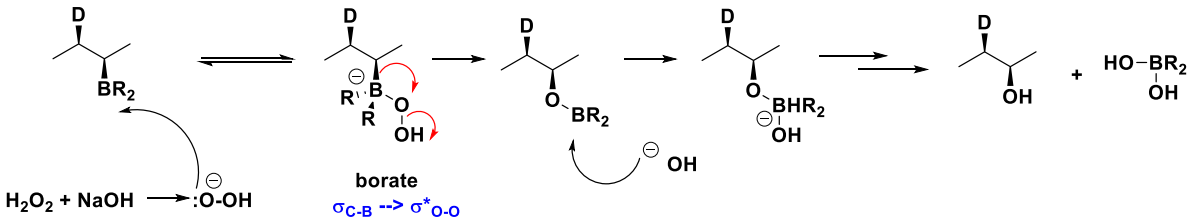
B_2H_6 BH_3 in THF 9-borabicyclo[3.3.1]nonane (9-BBN)

^nBu	94%	% of anti-Markonikov	99%	% of anti-Markonikov
	57%	% of anti-Markonikov	99%	% of anti-Markonikov
Ph	80%	% of anti-Markonikov	99%	% of anti-Markonikov

- **Hydroboration reaction is stereospecific. The oxidation reaction occurs with the retention of configuration.**

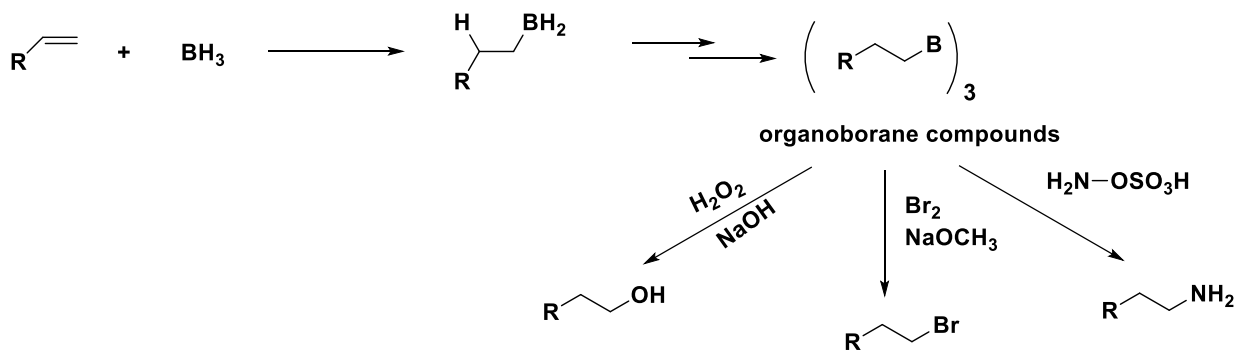


Mechanism of oxidation reaction

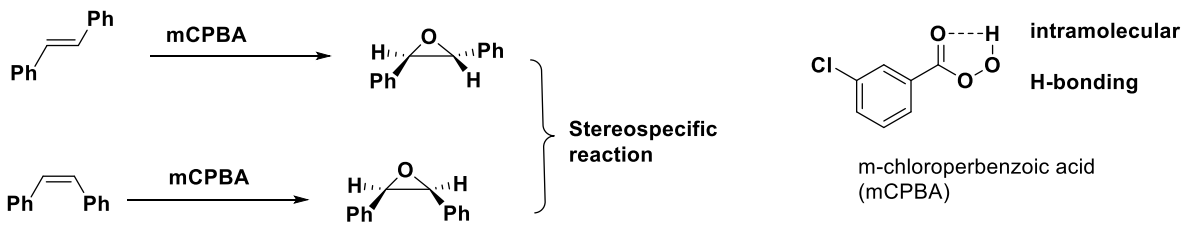


Class 4

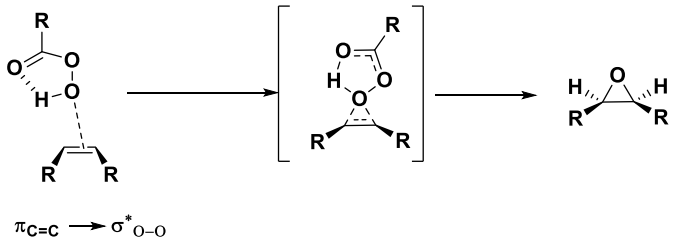
Applications of the organoborane compounds



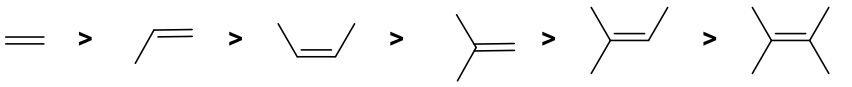
Epoxidation Reaction



Mechanism

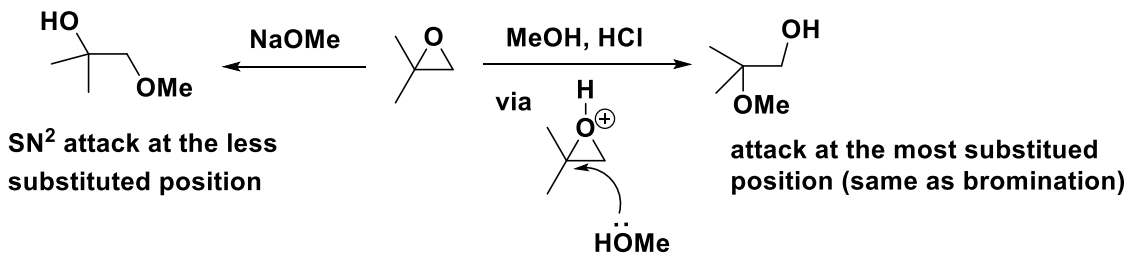


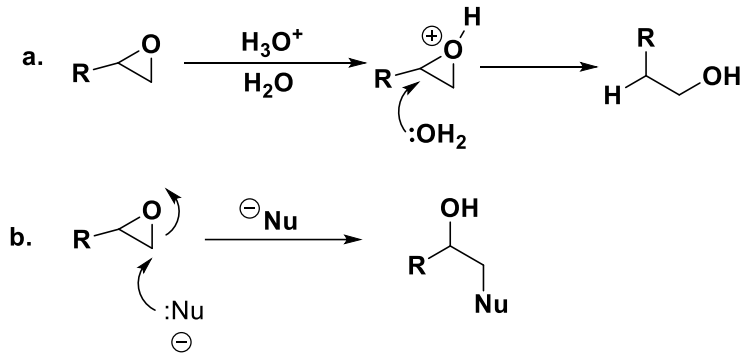
Rates :



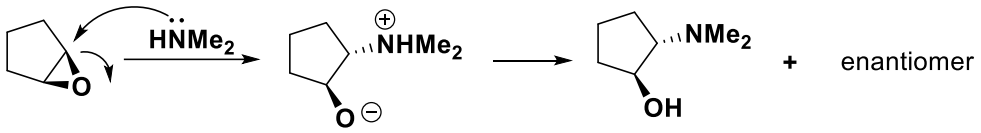
Class 4

Regioselectivity of epoxide opening





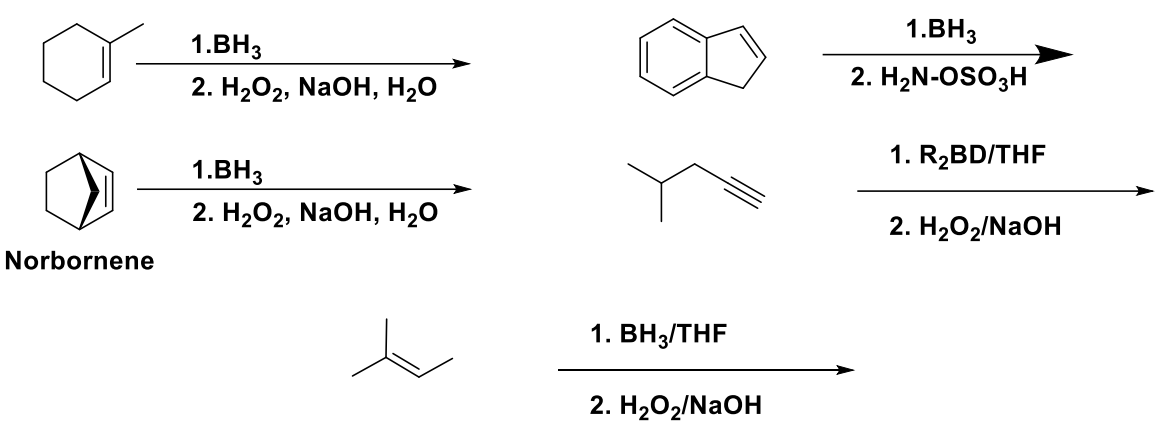
The reaction is stereoselective



Class 4

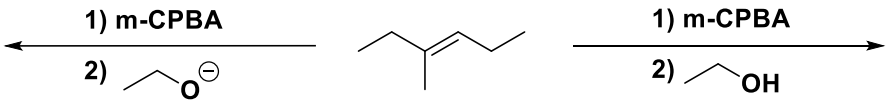
Questions

1.

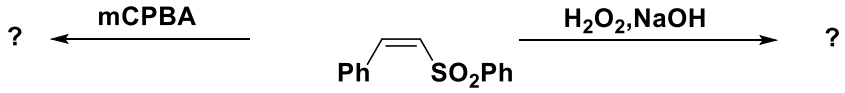


2. Write the products with proper mechanisms and stereochemistry.

a.



b.



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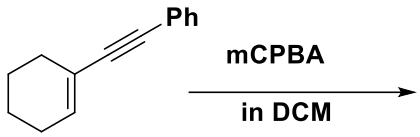
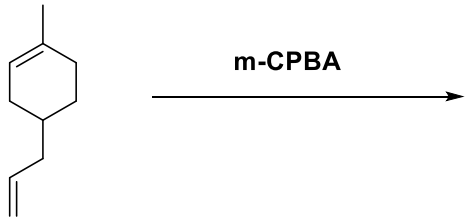
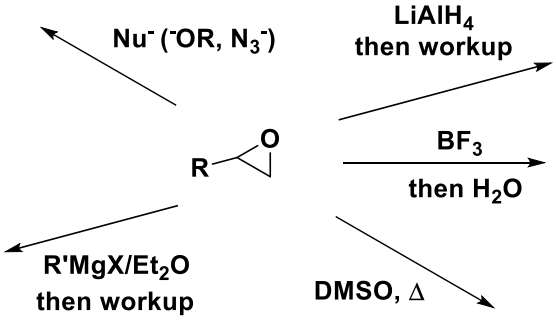


d.



3.